



DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

University Examination 2025/2026

**THIRD YEAR SECOND SEMESTER EXAMINATION FOR THE DEGREE OF
BACHELOR OF SCIENCE MECHATRONIC ENGINEERING**

5TH FEB.2026 EMT 3202: MECHATRONIC SYSTEM PROGRAMMING II 8.30AM-10.30AM

DATE: FEBRUARY 2026

TIME: 2 HOURS

Instructions

1. This paper contains FIVE questions
2. Answer Question **ONE** (Compulsory Question), **One** question in **Section A** (Question TWO or Question THREE) and **One** question in **Section B** (Question FOUR or Question FIVE)
3. Remember to indicate all steps and the relevant units in any calculations

Question ONE (Compulsory) [30 marks]

- a) An actuator **power source** is a key element in the selection of the correct actuator for an embedded system. Highlight the 3 main category of power sources and their common application. **[6 marks]**
- b) Using a well labelled block diagram, discuss the difference between **Synchronous and Asynchronous** types of communication in microcontrollers. **[6 marks]**
- c) As the lead project engineer you are to implement a **Weighted Average Filter** suitable for a low memory system. Explain how such a filter works and using the template below, write a C program function that implements such a filter for a given sensor input. **[5 Marks]**

Code Template

```
/* Filter Variables */
const float alpha = 0.1; // Smoothing factor
float smoothedValue = 0.0;
/* Filter Function */
float weightedAverageFilter (float raw_sensor_value) {
    /* Write your implementation Code here */
    return smoothedValue;
}
/* Filter Variables */
```

- d) Using suitable circuit diagrams discuss the difference between 8 bit and 4-bit LCD microcontroller operation and highlight the main advantage and disadvantage of 4-bit mode. [8 Marks]
- e) **Universal Asynchronous Receiver/Transmitter (UART)** is a common data communication method in embedded systems. Briefly, describe its functionality and explain the constituent parts of a typical UART data frame. [5 Marks]

SECTION A

Question TWO [20 marks]

- a) **Keypads** are very often used for User Interfacing (UI) in most embedded mechatronic systems. To easily interface with keypads, keypad encoders such as **MM74C922** are used. Explain how such encoders enable microcontrollers to distinguish between different keypad buttons. [6 Marks]
- b) Using a suitable diagram explain the principle behind using resistors to interface a 4x4 keypad to a microcontroller using a single analogue input (**1-wire keypad interfacing**). [6 Marks]
- c) With the aid of a diagram pulse wave diagrams, discuss how a **2-channel motor encoder** can be used to detect clockwise and anticlockwise rotation. [8 Marks]

Question THREE [20 Marks]

- a) **I2C**, also known as **two wire interfacing (TWI)**, is a widely adopted communication protocol of embedded systems. Using a suitable block diagram, describe the device connection and highlight 4 key advantages and disadvantages of this protocol. [6 Marks]
- b) Using a suitable block diagram describe inner structure of a **servo motor** and typical working principle involved in making it move. [4 Marks]
- c) You have been asked to design a servo motor control system which takes a reading from a **12 bit Analogue to Digital converter (ADC)** and operates a **servo motor from -90 degrees to +90 degrees**. The minimum control pulse of the servo motor is a **PWM signal with an ON time of 0.5ms and a maximum ON time of 1.5 ms**. The microcontroller system operates at 16MHz, operating an 8-bit timer to generate the PWM pulse. The timer prescaler values from the data sheet are 1, 8, 64, 256 and 1024. Write a suitable equation that can be used to convert the ADC input to the appropriate PWM count value. [8 Marks]
- d) Define what is meant by **Step angle** with regards to **stepper motors** and highlight how the step angle value affects the motion of the stepper motor. [2 Marks]

SECTION B

Question FOUR [20 Marks]

- a) **Variable choice** and **Look-up** tables are techniques used in writing efficient embedded code. With suitable examples, explain how these methods are applicable in optimizing embedded code for Memory efficiency and computational efficiency [6 Marks]

- b) You have been provided with the following datasheet specifications for an off-grid data recording device.

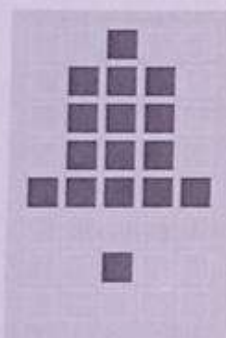
Microprocessor - Active mode current: **10mA**, Sleep mode current: **1mA**

Battery - **9 Volts**, with **200mAh** capacity

Firmware - 10 sec cycle: **Active mode for 1 second** and **sleep mode for 9 seconds** per cycle.

Using this data compute the **Average Current Draw** and the **Estimated Operation Time** for the system. [4 Marks]

- c) The custom character shown in **figure Q4b** is to be displayed on an LCD screen for an alarm clock application. Given the following LCD instruction set and assuming the following LCD commands are already stored in a separate header file, perform the necessary calculations to convert the custom character to hexadecimal values and write an executable C program using the template provided to load the values into the LCD CGRAM and displayed on the first row and fourth column of the LCD display. [10 Marks]



Existing functions:

Void LCD_Cmd(); A function used to send commands to the LCD processor

Void LCD_Char(); A function used to display characters to the LCD display

Void LCD_Init(); A function used to perform the necessary setup on the LCD display

Void LCD_Clear(); A function used to clear all the characters from the LCD display.

Figure Q4b: showing custom bell icon

LCD Commands:

0x04 – Set LCD CGRAM address

0x80 – Cursor set to the first row, the first column

0xC0 – Cursor set to the second row, the first column

Code Template

```
/*Typical Header file inclusion and CPU frequency definition*/
#define F_CPU 8000000UL
#include <avr/io.h>
#include <util/delay.h>
#include "LCD_Code.h" /*Inclusion of our own defined header file*/

/*Array to store hexadecimal values for custom characters */

/*Function Definition: To load the hex data into the first space in CGRAM*/
void Load_Custom_Char()
{
}

/*Main program*/
int main(void)
{
    return 0;
}
```

Question Five [20 Marks]

- a) Typical circuit implementation of a Stepper motor control circuit is as shown in the circuit **Figure Q5a**. The stepper motor has a step angle of 90 degrees and is to be controlled using half stepping. It is connected to a stepper motor driver IC ULN2003A through PORTB i.e. PB0 ~ PB3. The stepper motor is to be driven at 150 RPM. You have been contracted to write an executable control program to run the stepper motor.

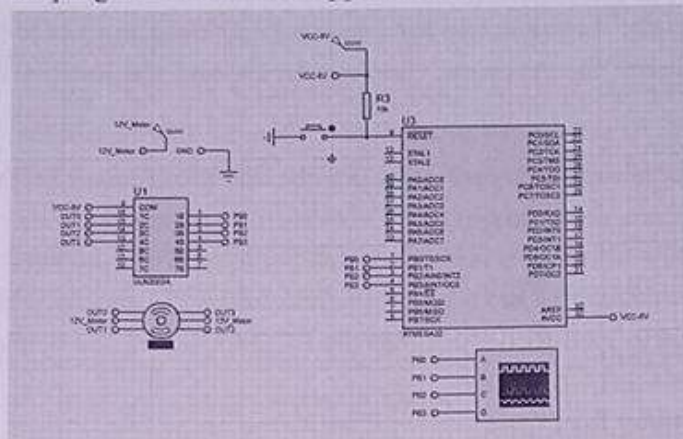


Figure Q5a: Stepper Motor Control Circuit

- Calculate the required motor step HEX values to be used on PORTB and present them in form of a table. **[6 marks]**
- Create an appropriate flow chart. **[4 marks]**
- Write the executable C program using the code template provided. **[4 marks]**

Show all calculations involved in the implementation of your system.

Code Template

```
#define F_CPU 8000000UL /* Define CPU Frequency 8MHz */
#include <avr/io.h> /* Include AVR std. library file */
#include <util/delay.h> /* Include delay header file */
/* Definition of the motor DDR and PORT */

#define Motor_Dir DDRB
#define Motor_Port PORTB

/* Hex values of the stepper motor step values */
int main(void)
{
    /* Motor port initialization */
    /* Time gap between each motor step */

    while (1) { /* Main motor rotation program */
        }
    return 0;
}
```

- b) Highlight the key features of **Serial Peripheral Interface** Communication Protocol using a suitable diagram to illustrate the connections between devices **[6 Marks]**